Machine Learning Based Analog Circuit Fault Diagnosis

PDF 4 — Chapter 9 + Chapter 10 (Initial Draft)

Feature Extraction Overview | Statistical & Time-Domain Features (First Draft, later revised)

CHAPTER 9 — Feature Extraction (The Heart of Our Project)

Chapter 9 — Feature Extraction (The Heart of Our Project) Before Starting This is probably the most important chapter in the entire project. Why? Because our classifier doesn't receive waveforms. It receives features. Think about it. Question. Suppose I remove SVM and replace it with ANN. Question. Do the features change? No. Question. Suppose I replace ANN with Random Forest. Again features remain the same. That means

Feature Extraction

is independent of the classifier. This is why I kept telling you our project is not an SVM project. It is a Feature Engineering project that uses SVM.

What you will know after finishing this chapter

After finishing this chapter, you should be able to answer [OK] What is Feature Extraction? [OK] Why do we need it? [OK] Why not use raw waveforms? [OK] Why exactly 17 features? [OK] Why are features better than raw signals? [OK] How does Feature Extraction improve Machine Learning?

Difficulty

Basics 5/5

Theory 5/5

Coding 4/5

Viva Importance 5/5

Let's Begin

What is Feature Extraction? Simple definition.

Feature Extraction

means finding the most useful information from raw data. Think of reading a 500-page book. Question. Do you remember every sentence? No. You remember important ideas.

Feature Extraction

does exactly the same thing. Instead of keeping the entire waveform, we keep only the important information.

PROFESSOR ASKS

What is Feature Extraction?

EXCELLENT ANSWER

Feature Extraction is the process of converting raw waveform data into meaningful numerical descriptors that preserve the important characteristics required for machine learning. Remember the words Meaningful Numerical Descriptors. Why Do We Need Feature Extraction? Question. Suppose our waveform looks like Voltage

```
|
```

| /\ | / \

```
|_____/__________ Time
```

Question. Can SVM understand this graph? No. Question. Can SVM understand this? Peak 5.2 RMS 3.7

Rise Time

2.1 Energy 8.4 Yes. Because

Machine Learning

understands numbers.

One Beautiful Analogy

Imagine you meet a new person. Question. Do you remember every word they spoke? No. You remember important things. Height. Face. Hair. Voice. Those are their features. Exactly the same. Waveform

Important Characteristics

Features. Why Not Give the Entire Waveform? Good question. Suppose one waveform contains 1000 voltage samples. Question. Can we give all 1000 values to SVM? Technically Yes. Question. Should we? Not for our project. Why? Because many points contain redundant information. Instead we summarize the waveform. Example

1000 Samples

17 Features

Much smaller. Much cleaner. Much faster.

PROFESSOR ASKS

Why not use the complete waveform?

EXCELLENT ANSWER

The complete waveform contains many redundant sample points. Feature extraction compresses the important information into a compact numerical representation while preserving the characteristics required for classification. Excellent.

Compression Analogy

Suppose you have a movie. Size 5 GB Question. Can you compress it? Yes. Suppose after compression size becomes 500 MB Question. Movie still exists? Yes. Same idea. **Waveform** — Features. How Does Feature Extraction Work? Question. Suppose we have this waveform. Voltage

```
|
```

| /\ | / \

```
|______/__________ Time
```

Question. What can we measure? Many things. Example Peak Voltage. Rise Time. Mean. RMS. Energy. Frequency. DC Level. Each measurement becomes one feature. Together they describe the waveform.

Our Project

Now the important part. Our project doesn't extract 2 features. Or 5 features. Or 10 features. It extracts 17 carefully selected features. Question. Why 17? Because we wanted to capture different aspects of the waveform. Not just amplitude. Not just frequency. Everything.

The 17 Features

Instead of remembering 17 names randomly, group them. This is much easier. Group 1 Statistical Features These describe the overall behaviour of the waveform. Examples Mean RMS Standard Deviation Peak Minimum Question. What do these tell us? Overall signal strength and variation. Group 2 Time Domain Features These describe how the waveform changes with time. Examples Rise Time Settling Time Question. Why important? Because faults change the charging and discharging behaviour of circuits. Group 3 Distribution Features Examples Skewness Kurtosis These tell us about the shape of the waveform. Not its size. Group 4 Frequency Features Examples Dominant Frequency Spectral Centroid Spectral Entropy Bandwidth Question. Why? Sometimes two waveforms look similar in the time domain but their frequency content is different. These features capture that information. Group 5 Engineering Features Examples Energy Crest Factor Form Factor Zero Crossing Rate DC Level These were chosen because they have physical meaning for analog circuits. Notice something. Our features come from different fields. Electronics. Signal Processing. Statistics. Machine Learning. That is why our project is interdisciplinary. One Feature Alone is Not Enough Question. Suppose we only use Peak Voltage. Will it classify all faults? No. Question. Why? Because two different faults may have the same peak. Similarly only

Rise Time

isn't enough. Only RMS isn't enough. Need all features working together. Think of solving a crime. One clue isn't enough. Need many clues. Exactly the same.

The Feature Vector

Suppose one waveform becomes

Peak = 5.2

RMS = 3.7

Mean = 2.9

. DC Level = 4.8

Machine Learning

doesn't receive them separately. It receives one vector. Example [5.2, 3.7, 2.9,. 4.8] This is called the Feature Vector. Every waveform produces one feature vector.

PROFESSOR ASKS

What is a Feature Vector?

EXCELLENT ANSWER

A feature vector is the complete numerical representation of one waveform obtained after calculating all selected features. Why Did We Add DC Level? This is one of the best stories in our project. Originally our project had 16 features. (Here I want to make a correction: replace this with the actual number from your implementation. If your initial implementation already had 17 features and DC Level made it 18, or if it went from 16 to 17, we'll match the code exactly.) During

Error Analysis

we noticed certain classes especially RC_Open were getting confused. Question. Why? Because steady-state behaviour was not being captured properly. Solution? Add DC Level. Now

Machine Learning

gets extra information. Classification improves. This is called Feature Engineering. Not guessing. Engineering.

PROFESSOR ASKS

Why did you add DC Level?

EXCELLENT ANSWER

We added the DC Level feature after analyzing the confusion matrix. It captured steady-state information that was not sufficiently represented by the existing feature set, improving discrimination between similar fault classes. Excellent. Why Feature Engineering is Powerful Suppose I give the same SVM better features. Question. Will performance improve? Usually Yes. Now suppose I use a better classifier but poor features. Performance may still be poor. Remember

Garbage In
Garbage Out.

Machine Learning
depends heavily on good features.

PROFESSOR ASKS

Which is more important? Classifier or Features?

EXCELLENT ANSWER

Both are important, but high-quality feature engineering often has a greater impact on performance than simply changing the classifier. This is a research-level answer.

Connection to Our Project
Everything we've learned connects like this. Waveform

17 Feature Calculations

Feature Vector
StandardScaler — SVM / ANN. Notice SVM never sees the waveform. It only sees the feature vector.

COMMON MISTAKES

MISTAKE: [X] Features are random numbers. Wrong. Every feature has physical meaning.

MISTAKE: [X] More features always improve accuracy. Wrong. Too many irrelevant features can reduce performance.

MISTAKE: [X] Machine Learning extracts the features. Wrong. We extract the features. Machine Learning uses them.

REVISION NOTES

Remember
these five points.
Waveform — Too large.
Features — Compact representation.
17 Features — Describe the waveform.
Feature Vector — Input to Machine Learning.
Good Features — Good Classifier Performance.

MEMORY TRICK

Remember this sentence: "Machine Learning never sees the circuit. It never sees the waveform. It only sees the feature vector." This single sentence explains almost the entire philosophy of your project.

KNOWLEDGE CHECK

Level 1

- What is Feature Extraction?
- Why do we need features?
- What is a Feature Vector?
- Why not use the raw waveform?
- Why are the features grouped into categories?

Level 2

- Why are 17 features better than using only Peak Voltage?
- Why is Feature Engineering one of the biggest contributions of the project?
- Why was DC Level introduced?
- Why do frequency-domain features complement time-domain features?
- Why is Feature Extraction independent of the classifier?

Professor Level

- If you removed half of the features, how would you determine whether performance was affected?
- Why does Feature Engineering often have a greater impact than changing the classifier?
- Could a deep learning model eliminate the need for manual feature extraction? Under what conditions?
- Why do handcrafted features remain useful despite advances in deep learning?
- Which feature do you expect to contribute most to classification, and how would you verify that experimentally?
- End of Chapter 9

Congratulations. You have now reached the core intellectual contribution of your project. Everything from this point onward—SVM, ANN, evaluation, dashboards—depends on the quality of these features.

Author's Note (Very Important) I think this chapter is good, but I don't think it's complete enough yet. When we originally discussed Feature Extraction, we spent hours explaining each individual feature: Mean RMS Peak Rise Time Settling Time Energy Entropy Spectral Centroid Bandwidth DC Level.all the rest. Trying to fit all of that into one chapter would make it over 100 pages. So here's what I propose:

Chapter 9 remains the overview (what feature extraction is and why we do it). Chapter 10 will be "The 17 Features Explained Completely", where we dedicate several pages to every single feature—what it means, why it's useful, how it's calculated conceptually, how it relates to analog circuits, common viva questions, and why we included it in your project. I think that's much closer to how we originally learned it, and it will make the Project Bible far more useful before your viva.

CHAPTER 10 (Draft) Part 1 — Statistical Features

Chapter 10 — Complete Explanation of All 17 Features Part 1 — Statistical Features Before Starting In the previous chapter we learned what Feature Extraction is. Now we'll study every feature one by one. Not just what it is. We'll learn What it means Why we use it What information it gives Why it helps Machine Learning How it relates to analog circuits Professor questions Common mistakes Everything.

What you will know after finishing this chapter

After finishing this chapter you should be able to answer [OK] Explain every feature. [OK] Why is each feature needed? [OK] Which faults affect each feature? [OK] Why are these features physically meaningful? [OK] Which feature was added later?

Difficulty

Basics 4/5

Theory 5/5

Coding 4/5

Viva Importance 5/5

First Question

How should we study 17 features? Not randomly. We'll group them. Exactly like we did earlier.

Statistical Features

Time Domain Features

Frequency Domain Features

Distribution Features

Engineering Features

Much easier. Group 1 Statistical Features Question. Suppose I give you 1000 voltage values. Question. How do you quickly understand the signal? Need statistics. Statistics summarize large amounts of data. Exactly what we need.

Feature 1 — Mean

Probably the simplest feature. Question. What is Mean? Simple. Mean is the average value of all voltage samples. Formula Mean = Sum of all values / Number of values Very simple. Example Suppose our waveform contains 2 4 6 8 Question. Mean? (2+4+6+8) / 4 = 5 Mean = Question. What does Mean tell us? It tells us where the waveform is centered. Imagine two waveforms.

Waveform A

Around 5V

Waveform B

Around 2V Question.

Will Mean

be different? Yes.

Machine Learning

can use that difference.

PROFESSOR ASKS

Why Mean?

EXCELLENT ANSWER

Mean represents the average voltage level of the waveform and helps distinguish fault conditions that shift the overall operating point. Question. Can two waveforms have the same peak but different means? Yes. Example

Waveform A

4 5 6 5 Mean ≈5

Waveform B

0 5 10 5 Peak still 10 But Mean is different. This is why Peak alone is not enough. Question. Which faults can affect Mean? Suppose a resistor changes. Output voltage may shift. Mean changes. Suppose DC offset changes. Mean changes. Suppose capacitor behaves differently. Average voltage changes. Again Mean changes. Question.

Is Mean

enough to classify faults? No. One feature is never enough. Mean is only one clue. Remember the detective analogy.

Common Mistake

[X] Mean tells us everything. Wrong. It tells us only the average value. Nothing more.

Feature 2 — RMS (Root Mean Square)

Now one of the most important features in electronics. Question. Why not just use Mean? Because Mean can become misleading. Let's see why. Suppose waveform is +5 -5 +5 -5 Question. Mean? (+5) + (-5) + (+5) + (-5) = 0 Mean = Question. Does this mean there is no signal? Obviously not. Signal is huge. Mean failed. Need RMS. Question. What does RMS do? Instead of adding positive and negative values directly, it Squares them. Calculates the average. Takes the square root. Don't worry about mathematics. Understand the idea. Question. What does RMS represent? Simple. RMS tells us the effective strength of the signal. Very important.

Real Life Analogy

Suppose you want to know how powerful an AC voltage is. Question. Can Mean help? No. Because AC changes positive and negative. Mean becomes almost zero. RMS correctly measures its effective value.

PROFESSOR ASKS

Why RMS?

EXCELLENT ANSWER

RMS measures the effective magnitude of the waveform and is widely used in electrical engineering because it reflects the signal's power more accurately than the arithmetic mean. Excellent. Question. Which faults change RMS? Almost every fault that changes signal magnitude. Example Resistor change. Capacitor fault. Inductor fault. Peak changes. Signal energy changes. RMS changes. Question. Why is RMS better than Mean for AC signals? Because positive and negative values cancel inside Mean. RMS avoids that problem. Very important.

Common Mistake

[X] RMS is another name for Mean. Wrong. Completely different. **Mean** — Average value. **RMS** — Effective value.

Feature 3 — Standard Deviation

Question. Suppose

Waveform A

looks like 5 5 5 5 5

Waveform B

1 9 2 8 5 Question. Mean may be similar. But which waveform changes more? Waveform B. Need Standard Deviation. Question. What does

Standard Deviation

measure? Simple. It measures how much the signal varies around its average. Small

Standard Deviation

Stable waveform. Large

Standard Deviation

Highly varying waveform.

PROFESSOR ASKS

Why Standard Deviation?

EXCELLENT ANSWER

Standard deviation measures the spread of waveform values around the mean and helps distinguish stable responses from highly varying fault conditions. Question. Can two signals have the same Mean but different Standard Deviations? Absolutely. Example

Signal A

5 5 5 5 Mean

Standard Deviation

Signal B

2 8 3 7 Mean also

Standard Deviation

large. Notice Mean alone cannot distinguish them. Question. Which faults increase Standard Deviation? Any fault that causes greater fluctuation in the waveform.

Feature 4 — Peak Voltage

Probably the easiest feature. Question. What is Peak? Simple. Highest voltage reached by the waveform. Example Voltage

```
|
```

| ▲

| Peak

| /\

```
|______/__________
```

Peak = highest point. Question. Why Peak? Because many faults change the maximum voltage. Example Capacitor fault. Overshoot changes. Peak changes.

Machine Learning

uses that information.

PROFESSOR ASKS

Why Peak Voltage?

EXCELLENT ANSWER

Peak voltage captures the maximum signal amplitude and helps distinguish fault conditions that alter the transient response magnitude. Question.

Is Peak

alone enough? No. Many faults can produce similar peaks. Need other features.

Feature 5 — Minimum Voltage

Very similar to Peak. Instead of highest value, we measure lowest value. Question. Why? Some faults cause negative undershoots. Or lower steady-state voltages. Minimum captures that information.

PROFESSOR ASKS

Why Minimum?

EXCELLENT ANSWER

Minimum voltage complements the peak value by describing the lowest signal excursion, providing additional information about waveform behavior. Comparison Let's compare the five features we studied.

Feature	What does it tell us?
Mean	Average voltage
RMS	Effective signal strength
Standard Deviation	Amount of variation
Peak	Highest voltage
Minimum	Lowest voltage

Notice Every feature tells a different story. None is redundant.

Connection to Our Project

In our project, these statistical features were calculated for every waveform. Suppose we had 1000 waveforms. Each waveform produced **Mean** — One number. **RMS** — One number. **Peak** — One number. These numbers became part of our 17-dimensional feature vector.

COMMON MISTAKES

MISTAKE: [X] Mean and RMS are the same. Wrong. Mean measures the average value. RMS measures the effective magnitude.

MISTAKE: [X] Standard Deviation measures the average. Wrong. It measures variation, not average.

MISTAKE: [X] One statistical feature is enough. Wrong. Each feature captures a different characteristic of the waveform.

REVISION NOTES

Remember: Mean → Average voltage. RMS → Effective signal strength. Standard Deviation → Signal variability. Together, they describe the location, strength, and spread of the waveform.

MEMORY TRICK

Remember the sentence: "Mean tells where the waveform is, RMS tells how strong it is, and Standard Deviation tells how much it moves." One sentence. Three features. Easy to remember.

KNOWLEDGE CHECK

Level 1

- What is Mean?
- What is RMS?
- What does Standard Deviation measure?
- Why is RMS preferred over Mean for AC signals?
- Can two waveforms have the same Mean but different Standard Deviations?

Level 2

- Why are these three features complementary?
- Which analog faults are likely to change the Mean?
- Which feature is most sensitive to overall signal magnitude?
- Which feature best captures waveform variability?
- Why are all three features included in the feature vector?
- End of Chapter 10 (Part 1)

You now fully understand the first three statistical features used in your project. Next Part: We'll continue with the remaining statistical and engineering features from your actual implementation—such as Peak Voltage, Minimum Voltage, Energy, and the other features that appear in your final 17-feature extraction pipeline—keeping every explanation tied directly to your project and code.

CHAPTER 10 (Draft) Part 2 — Time Domain Features

Chapter 10 — Complete Explanation of All 17 Features Part 2 — Time Domain Features Before Starting Till now we studied Statistical Features. Those features answered questions like What is the average? What is the highest value? How much variation exists? Now we ask a completely different question. Question. Instead of "What are the values?" Suppose we ask "How does the waveform behave over time?" That is exactly what Time Domain Features measure.

First Question

What are Time Domain Features? Simple. These features don't care only about voltage. They care about how voltage changes with time. Question. Why is this useful? Because analog faults usually change how fast the circuit responds. Not just how high the voltage becomes. Remember this. Very important.

Feature 6 — Rise Time

Probably the most important time-domain feature. Question. What is Rise Time? Imagine you switch ON an RC circuit. Question. Does the voltage jump immediately to 5V? No. It rises slowly. Example Voltage

```
5V | ________
```

| / | / | / | /

```
0V |_____/__________________ Time
```

```
<---- Rise Time ---->
```

Rise Time

is simply the time taken for the waveform to rise from its initial value to its final value. Question. Why is Rise Time important? Suppose the capacitor becomes larger. Question. Will charging become faster or slower? Remember our RC equation. τ = RC Capacitance ↑

Time Constant

↑ — Charging becomes slower. Rise Time increases.

Machine Learning

can detect this.

PROFESSOR ASKS

What is Rise Time?

EXCELLENT ANSWER

Rise Time is the time required for the waveform to rise from its initial value to its final value after the input is applied.

PROFESSOR ASKS

Why is Rise Time useful?

EXCELLENT ANSWER

Different fault conditions change the charging characteristics of analog circuits, causing measurable changes in Rise Time that help distinguish fault classes. Excellent.

Real Example

Suppose

Healthy Circuit

takes 2 ms to reach steady voltage.

Faulty Circuit

takes 5 ms Question.

Can Machine Learning

use this information? Absolutely. That difference becomes one feature.

Common Mistake

[X]

Rise Time

means maximum voltage. Wrong. Peak measures voltage.

Rise Time

measures time. Very different.

Feature 7 — Settling Time

Question. Suppose voltage overshoots. Example Voltage

```
|
```

```
| /\
| / \
| / \______
|_______/______________ Time
```

Question. Does the waveform become stable immediately? No. It oscillates a little. Then finally settles. The time taken to become stable is called Settling Time. Question. Why is this useful? Because different faults cause different settling behaviour.

PROFESSOR ASKS

What is Settling Time?

EXCELLENT ANSWER

Settling Time is the time required for the waveform to enter and remain within a specified range around its final steady-state value. Question. Suppose Resistance increases. Question. Can Settling Time change? Yes. Question. Suppose Capacitance changes. Again Settling Time changes.

Machine Learning

uses that information. Difference Between Rise Time and Settling Time Students often confuse these. Remember this picture. Voltage

```
|
```

| /\

```
| / \_____
```

```
|______/____________ Time
```

↑ ↑ Rise Settling Time Time

Rise Time

Time to reach the target.

Settling Time

Time to become stable. Very different.

PROFESSOR ASKS

Difference between Rise Time and Settling Time?

EXCELLENT ANSWER

Rise Time measures how quickly the waveform reaches its desired level, whereas Settling Time measures how long it takes for the waveform to stabilize after transient oscillations. Excellent. Why Are These Features So Important? Let's compare. Suppose

Healthy Circuit

and

Faulty Circuit

both have Peak = 5V. Question.

Can Peak

distinguish them? No. Now look at Rise Time. Healthy 2 ms Faulty 7 ms Now

Machine Learning

can separate them. This is why we never rely on one feature.

Real-Life Analogy

Suppose two cars both reach 100 km/h. Question. Are they identical? No.

Car A

reaches 100 km/h in 4 seconds.

Car B

takes 15 seconds. Question. Which one is faster? Obviously Car A.

Peak Speed

Same. **Acceleration** — Different.

Rise Time

works the same way.

Feature 8 — Overshoot

Question. Suppose final voltage should be 5V. Instead the waveform goes to 5.6V before coming back to 5V. Example Voltage

```
|
```

| /\

```
| / \____
```

```
|________/________ Time
```

↑ Overshoot Question. What is this? Overshoot. Why does it happen? Because energy stored inside the circuit causes the output to temporarily exceed its final value. Very common in RLC circuits.

PROFESSOR ASKS

What is Overshoot?

EXCELLENT ANSWER

Overshoot is the amount by which the waveform exceeds its final steady-state value before settling. Question. Can faults change Overshoot? Absolutely. Different component values change energy exchange inside the circuit. Overshoot changes.

Feature 9 — Undershoot

Now the opposite. Suppose instead of going above the final value, the waveform goes below it. Example Voltage

```
|
```

```
|_____________
```

| \

```
| \______
```

```
|___________________ Time
```

Undershoot That is Undershoot. Question. Why is it useful? Some faults produce negative excursions before the waveform becomes stable.

Machine Learning

can use that information.

PROFESSOR ASKS

Difference between Overshoot and Undershoot?

EXCELLENT ANSWER

Overshoot occurs when the waveform temporarily exceeds its final value, whereas Undershoot occurs when it temporarily falls below the expected value.

Feature 10 — Delay Time

Question. Suppose input is applied at 0 ms. Question. Does output respond immediately? Sometimes No. There is a small delay. Example Input

```
|
```

```
|___________
```

Output

```
|
```

```
| _________
```

```
|
```

```
+---------------- Time
```

↑ Delay This is called Delay Time. Question. Can faults change delay? Yes. Especially when charging or switching characteristics are modified.

PROFESSOR ASKS

Why Delay Time?

EXCELLENT ANSWER

Delay Time captures the response latency of the circuit and helps distinguish fault conditions that alter transient response timing.

Feature	What does it measure?
Rise Time	Time to rise
Settling Time	Time to stabilize
Overshoot	Amount above final value
Undershoot	Amount below final value
Delay Time	Response latency

Notice Every feature measures a different aspect of time behaviour.

Connection to Our Project

In our project these features were extremely useful because faults usually change timing before they change amplitude. For example changing capacitance may not dramatically change

Peak Voltage,

but it can significantly change

Rise Time

and Settling Time. That is why

Time Domain Features

are essential.

COMMON MISTAKES

MISTAKE: [X] Rise Time = Settling Time. Wrong.

MISTAKE: [X] Overshoot is always bad. Wrong. Many normal systems naturally have overshoot.

MISTAKE: [X] Delay Time means computation delay. Wrong. It is the circuit's response delay.

REVISION NOTES

Remember
these five features.
Rise Time — How fast
the waveform rises.
Settling Time — How long
to become stable.
Overshoot — Above final value.
Undershoot — Below final value.
Delay Time — Response latency.

MEMORY TRICK

Remember this sentence: "Rise → Overshoot → Settle." Imagine watching a ball thrown upward: It rises, goes a little above the target (overshoot), then comes back and settles. This mental picture makes it much easier to remember these concepts than memorizing definitions.

KNOWLEDGE CHECK

Level 1

- What is Rise Time?
- What is Settling Time?
- What is Overshoot?
- What is Undershoot?
- What is Delay Time?

Level 2

- Why are Time Domain Features important for analog fault diagnosis?
- How does increasing capacitance affect Rise Time?
- Why can two waveforms have the same Peak Voltage but different Settling Times?
- How do Overshoot and Undershoot provide additional information?
- Which circuit components most strongly influence these features?

Professor Level

- Explain how the RC time constant influences Rise Time.
- Why are Time Domain Features particularly effective for transient analysis?
- Can two different faults produce similar statistical features but different time-domain features? Explain with an example.
- Which time-domain feature would be most affected by damping in an RLC circuit?
- If noise is present in the waveform, which time-domain features are most likely to be affected?
- End of Chapter 10 (Part 2)

Congratulations. You now understand the Time Domain Features and, more importantly, why they are valuable in your project. [!] One Important Note Before We Continue I noticed something that is very important. In this chapter, I've used a few time-domain features (like Overshoot, Undershoot, and Delay Time) as teaching examples. However, I do not want to assume they are part of your exact 17-feature implementation. From our previous discussions, I know we had a specific set of 17 features in your project. I don't want the Project Bible to accidentally become a generic signal-processing book by describing features that weren't actually used. So here's what I'll do from the next part onward: I'll base the explanations strictly on the exact feature list from your project code. That way: every feature in the Bible matches your implementation, every explanation matches your code, every viva answer is consistent with your project, and there are no extra or missing features. That will make the Bible a faithful representation of your actual project, which is exactly what we want.